Rafik Mhadhbi

Full-Stack Engineer · AI agents & real-time voice

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Summary

Full-stack engineer who takes AI agents from demo to production — chat and voice — across TypeScript, Python and Go.

Two and a half years owning production systems at a hospitality AI company: a guest-facing LLM agent on hundreds of hotels, a Go/WebRTC/SIP voice gateway built from zero, and a multi-tenant NestJS/MongoDB platform in 36+ languages.

Open to full-stack or backend roles where AI agents, real-time systems or multi-tenant platforms meet production. Remote or hybrid; three-month notice.

Experience

Quicktext

Jun 2024 – Present

Full-Stack Engineer — AI agents & voice

Sousse, Tunisia (hybrid)

`Platform & data at fleet scale`

- Built the LLM-facing layer of a NestJS/MongoDB platform serving thousands of hotel tenants — generative search with dependency resolution, Redis-cached settings sync, queued fan-out.
- Ran fleet-scale data work safely — bulk migrations across thousands of tenants with dry runs and snapshots, a multi-million-row warehouse backfill, repairs reconciled across three data stores.
- Owned release engineering on Kubernetes/gitops — multi-repository release cuts, release gating by revert, and git as the single source of truth for cluster configuration.

`AI agents in production`

- Own the production runtime of a guest-facing LLM agent (Python, FastAPI, Gemini) live on hundreds of hotels — trace-driven root causes, deterministic guardrails over prompt rules, QA sweeps of 300+ bugs.
- Built evaluation and cost tooling — a 219-case replay suite, LLM cost dashboards reconciled to the cloud bill, and an Elasticsearch tracer tuned from 21 s to 2 s per query.
- Designed the LLM-facing data layer — per-field descriptions and templates that cut hallucinations, plus an MCP server for validated read/write access to hotel configuration.

`Real-time voice & telephony`

- Designed and built from zero a Go media gateway (Pion WebRTC, SIP, Opus) carrying WhatsApp and phone calls into the AI agent — from ~12 to hundreds of concurrent calls per instance, measured.
- Connected telephony end to end — Twilio numbers and hotel switchboards over SIP (per-hotel digest auth), department transfers, links by SMS, and a speech-to-speech runtime sharing memory with chat.
- Secured the voice stack — provider CIDR allow-lists, per-switchboard digest auth with lockouts, and a token-signed voice endpoint rolled out without breaking deployed widgets.

ReBootKamp (RBK) — JavaScript Instructor

Oct 2023 – May 2024

Taught full-stack JavaScript (React, Node.js, SQL and NoSQL) and mentored student teams through project delivery.

E-Tafakna — Full-Stack Developer (internship)

Sep 2023

Built phone and e-mail verification end to end and contract search and navigation for a React Native app with an Express backend.

Skills

Build the agent	Python · FastAPI · LLM agents (Gemini, Claude) · Prompt & guardrail design · Evaluation harnesses · MCP · RAG / pgvector · Elasticsearch tracing
Carry the voice	Go · WebRTC (Pion) · SIP · RTP / Opus · Twilio Voice & SMS · WhatsApp Business Calling · Realtime speech-to-speech
Run the platform	TypeScript · Node.js · NestJS · MongoDB · PostgreSQL · Redis / BullMQ · Kubernetes · gitops & CI · Docker · Datadog / Sentry · Laravel (legacy)
Ship the product	React · Next.js · Astro · Tailwind · i18n / ICU (36+ languages) · Accessibility · REST / OpenAPI · Zod · Vitest / Playwright

Independent projects

RFPOffice — Co-founder, technical lead (part-time)

2026 – Present

AI SaaS turning RFP documents into proposals — NestJS 11 API with 12 BullMQ queues, Supabase Postgres + pgvector RAG, Claude for drafting and Gemini embeddings, OpenAPI-generated client, token-cost ledger; TanStack Start front end.

Moved production data (19 tables, 280+ files) off a no-code backend in one day; RAG evaluation (hit@K, MRR) runs as a CI gate.

Tunigoo — Founder (part-time)

2026 – Present

Airport-transfer and car-rental platform — Prisma/PostgreSQL API, prerendered React client in French, English and Arabic, multi-currency pricing with daily FX refresh, resumable distance measurement via OpenRouteService.

Education & certification

Full Stack JavaScript Developer (certificate) — ReBootKamp, Mar – Sep 2023

Micro Computer Systems Technician (diploma) — CSFM Nabeul, 2021 – 2023

Languages

Arabic (native) · French (professional) · English (professional)